The University of Sydney

BIOL2022
Biology Experimental Design and Analysis

BEDA

Unit Handbook

Dr. Januar Harianto
Prof. Clare McArthur
Prof. Matthew Crowther

Semester 2 · 2026

Third edition · Semester 2, 2026

About this edition

This edition of the *Biology Experimental Design and Analysis Unit Handbook* was prepared for BIOL2022 at the University of Sydney.

Biology Experimental Design and Analysis changes as we refine the teaching, practical activities, datasets and assessments. Students returning to this material should use the edition from the year in which they completed the unit. Later editions may differ, sometimes substantially.

Authors

Januar Harianto · Clare McArthur · Matthew Crowther

How this book was built

This handbook is written in Quarto using Markdown and R. The online handbook is published as HTML, while the A4 edition is typeset with Typst. The print edition uses Libertinus Serif throughout. The source is developed openly on GitHub, and meaningful changes are recorded on the Updates page.

Rendering the handbook

Most students should use the published PDF. Students who want to build a local copy can install Quarto 1.9 or later and R. The rendering instructions in the source repository explain the remaining steps.

Licence

Online handbook: https://biol2022.github.io/BEDA-handbook

Source: https://github.com/BIOL2022/BEDA-handbook

Updates: https://biol2022.github.io/BEDA-handbook/updates.html

Suggested citation

Harianto, J., McArthur, C., & Crowther, M. (2026). *Biology Experimental Design and Analysis Unit Handbook* (3rd ed.). https://biol2022.github.io/BEDA-handbook

This handbook has been shaped by the students and staff who have studied and taught Biology Experimental Design and Analysis. Their questions, feedback and practical experience continue to inform each edition.

Contents

1

Unit information

1.1 Welcome

Thank you for joining **BIOL2022: Biology Experimental Design and Analysis**, also known as BEDA. The ability to critically evaluate evidence is a fundamental life skill. In BEDA, we build this skill through hands-on work in experimental design and data analysis. How do we design a robust study? How does a study's design shape its results? How do we choose statistical tools that fit the research question? The reasoning skills you develop will be invaluable not just in biology, but in *any* future career path you choose.

While the focus of BEDA is on the reasoning behind data analysis and interpreting results, we also provide practical guidance on using software. We recommend **Jamovi** if you are new to statistical software. If you already have experience with **R**, you are welcome to use it. **SPSS** and PRIMER are also supported but we prefer to use open-source software where possible as they remain accessible to all.

1.2 Assumed knowledge

BEDA is designed to build on introductory statistics (rather than teach it from the beginning). We use ideas such as data types, sampling variation, hypothesis testing, p-values, confidence intervals and simple statistical models from the opening weeks. This allows us to spend more time on designing biological studies, building appropriate models and interpreting evidence.

Because BEDA has broad prerequisites, qualifying for enrolment into this unit does **not** always mean that you have studied statistics. **However, taking a first-year statistics unit is an assumed knowledge for BIOL2022**. Not fully meeting the assumed knowledge may make BEDA *extremely* challenging as we have limited time to explain basic concepts that are assumed to be known from first-year statistics.

Some enthusiastic students choose to take BEDA straight after their first semester of first year. If this is you, there may be extra challenges, as report writing, group work, and time management skills are usually developed across the full first year of university. For this reason, we strongly recommend taking BEDA in your second year (as intended), so you are better prepared for the workload and expectations of a unit designed for students with more experience.

If you decide to stick around, use Am I ready for BEDA? to see what you need to review so that you can get the most out of the unit.

1.3 Modules

BEDA is organised into three modules, each taught by a different lecturer. Ideas introduced early in the unit continue through the semester as they are applied to different kinds of biological research. Importantly the focus is on the reasoning behind the analysis and how the data are collected, rather than just the analysis itself.

1.3.0.1 Module 1 (Weeks 1–5)

Januar introduces the foundations of study design and statistical modelling. You will look at how a research question, the way a study is designed and the data you collect come together in a model. Some students may be familiar with these ideas from first-year statistics, but we will explore them in a new context. **Importantly, we will use this module to ensure that all students are on the same page with the foundations of study design and statistical modelling, so that we can build on these ideas in later modules**.

The module will introduce you to the following familiar ideas in a new context:

- sampling and how it affects the conclusions you can draw
- study design and variables
- graphical models
- the general linear model (GLM)
- model assumptions of the GLM
- transformations and why we use them
- control variables and random effects as opposed to procedural controls

These ideas help you begin designing your experiment for Report 1.

1.3.0.2 Module 2 (Weeks 6–8)

Clare develops the modelling workflow through univariate experiments, where the analysis focuses on one response variable. You will examine experiments involving categorical predictors, continuous relationships and binary responses, then consider how to choose between possible models.

The module will see your starting to collect data for Report 1. You will also be coached on how to write a report that clearly communicates your study design, analysis and interpretation of results. Additional concepts introduced in this module include:

- binary responses
- non-normal data and dealing with it
- picking the right model and choosing between models

1.3.0.3 Module 3 (Weeks 9–12)

Mat extends the same reasoning to multivariate data, where several response variables are considered together. You will learn how ordination methods such as principal component analysis and multidimensional scaling can reveal patterns in complex datasets, how clustering identifies similar observations, and how multivariate tests evaluate evidence for group differences. **Importantly, you will learn that multivariate methods are great for exploring patterns that you see in the world around you (and are thus very popular in biology)**.

Report 2 brings these ideas together. You will design a multivariate study, collect a group dataset and use it to describe, test and interpret patterns in your environment. We will cover:

- the reason behind multivariate methods
- the ordination methods PCA and MDS
- clustering
- multivariate tests for group differences using PERMANOVA
- ANOSIM and SIMPER for post-hoc analysis

1.4 Lectures & pracs

There are two lectures each week. Unfortunately we do not get to pick the locations, so they are held in different buildings on different days. You are expected to attend 80% of lectures in one way or another. We record this attendance in-person and online when you watch the recordings.

- **Tuesday, 10–11 am** — Belinda Hutchinson Building, Lecture Theatre 1110
- **Wednesday, 10–11 am** — Sydney Nanoscience Hub, Lecture Theatre 4002 (Messel)

You will also attend one two-hour practical each week in your allocated class:

- **Tuesday, 2–4 pm or 4–6 pm**
- **Wednesday, 2–4 pm or 4–6 pm**
- **Friday, 2–4 pm or 4–6 pm**

All practicals are held in **Carslaw Building, Wet Lab 307**. Check your personal timetable for your allocated class and any timetable changes. Do note that because we are a large cohort, once you are allocated to a practical class, you cannot change it. To change your practical class, you must contact the unit coordinator and provide a valid reason, normally a time table clash with another unit or caring responsibilities.

Attendance: You are expected to participate in at least 80% of timetabled activities and must attend at least 10 of the 12 practicals. See **Attendance** under the Code of Conduct for further details.

1.5 Code of conduct

A code of conduct is a shared agreement about how we learn and work together.

BEDA should be a respectful and inclusive place to learn. Treat classmates and staff with courtesy, including when you disagree. Use people's correct names and pronouns, respect different backgrounds and levels of experience, make space for others to contribute, and avoid disrupting lectures or practicals. Discrimination, intimidation and harassment are not acceptable. Follow staff instructions and all laboratory safety requirements when working in laboratory spaces.

1.5.0.1 Working with your group and data

Contribute fairly to your group's work and raise participation problems early. As with any group work, the issues that arise are often best resolved by talking to your group members first. It is important to use a group charter to clarify expectations and responsibilities including how to share data and work together.

Share group data and the information needed to interpret it promptly. Never fabricate data, or omit or alter observations without documenting and justifying the decision. You may discuss your analysis with your group, but any individual report must be written and submitted independently.

More information on how to work work with your group is provided with the Report 1 and Report 2 instructions. If you are unsure about what is acceptable, ask your practical demonstrator or the unit coordinator.

1.5.0.2 Attendance

Attend lectures live where possible. If you miss a lecture, watch the recording promptly. You are expected to have attended or watched the Tuesday lecture before your practical. Because practicals run on Tuesday, Wednesday and Friday, the Wednesday lecture is not assumed knowledge for that week's practical, but you should attend or watch it during the same week.

You must attend at least 10 of the 12 practicals to meet the 80% attendance requirement. Practical attendance is not managed through special consideration. Record your own attendance during each practical using the QR code or website link, then check that it has been recorded correctly. You have one week to request a correction.

1.5.0.3 Raising a concern

For routine group-work, data-sharing or class concerns, speak with your practical demonstrator or the unit coordinator as early as possible. You do not need to approach the teaching team first if the matter is serious or you would be uncomfortable doing so. The University provides separate pathways for complaints, bullying, harassment and discrimination and confidential support through Safer Communities for sexual harm or gender-based violence. In an immediate emergency, call 000 and follow the University's emergency and campus safety guidance.

2

Week 1: Getting started

Biology Experimental Design and Analysis (BEDA)

Introduction

Welcome to your first BEDA practical. Imagine that this is the first day of your new job. A decent workplace environment would see you being introduced to the people, the tools and the resources you will use for your job. Maybe you have done it before, but there is always something new. This practical is *that* introduction.

This week, you will get familiar with the tools and approaches you will use throughout the semester. You will also have time to explore the software and support resources before completing your first modelling task, which is to communicate your ideas using a model, even when you do *not* yet have a statistical model to fit.

Before the practical

1. Attend or watch the Week 1, Tuesday lecture.
2. Read through this practical before class. You will have a better experience if you are familiar with the tasks. **You do not need to complete anything in advance, but reading ahead will help you plan your time and know what to expect.**
3. Bring a laptop. If possible, install your chosen software and the required modules or packages before the practical. If setup is not finished when class begins, work with a partner while the downloads continue.
4. Bring a pen or pencil for your lab notebook. You will use it to sketch ideas, annotate figures, and quickly record decisions during activities.

ⓘ **Note:** Please avoid asking AI to summarise this practical before you have read it yourself. You need to develop critical reasoning skills, which often requires engaging directly with the material and tasks. It takes about 5 minutes to read, and doing so will help you understand what you do (and do not) know so that you can ask for the right help during the practical.

i Download the Week 1 datasets

Download both files before you begin and keep their filenames unchanged:

- Download **week01-penguins.csv**
- Download **week01-possums.xlsx**

If the links do not work, you can also download the files from Canvas - Module Content and Resources.

Learning outcomes

By the end of this practical, you should be able to:

Learning outcome	How you will practise it
Formulate a **measurable** biological question from observations or available data.	Turn observations about penguins into questions, then formulate your own question using variables from another dataset on possums.
Distinguish response and predictor **variables** and classify them by data type.	Classify variables, form biologically meaningful pairs and identify what you want to explain and what might explain it.
Explain how decisions made during **study design** shape the models that can be used.	Compare age recorded as categories with age measured continuously, then consider how measuring or classifying your own variables differently would change the model.
Use a **graphical model** to communicate an expected biological relationship *before* analysing data.	Choose and justify an appropriate plot, label both axes and sketch a plausible relationship in your lab notebook.

2.1 Workshop

We will begin with a short workshop to help you get started.

2.1.1 Pick your software (15 min)

If you are new to statistical software, please choose Jamovi. Demonstrators with expertise in R and SPSS will also be available, although we recommend Jamovi if you would otherwise choose SPSS.

ⓘ **Note:** If you use other software, we assume that you are already familiar with it and can complete the practical tasks independently.

2.1.1.1 Jamovi

1. Go to the Jamovi download page.
2. Download the current **Desktop** version for your operating system.
3. Open the downloaded installer and follow the prompts. On macOS, open the `.dmg` file and move Jamovi to your Applications folder.
4. Open Jamovi.
5. Click the + icon in the top-right corner and select **jamovi library**.
6. Select **Available**, then install **GAMLj3**, **vijPlots** and **flexplot**.

2.1.1.2 R/RStudio

R and RStudio are separate programs. **Install R first, then install RStudio.**

1. Download R for Windows or macOS. On a Mac, choose the installer that matches Apple silicon or Intel.
2. Open the R installer and accept the default options.
3. Download the free version of RStudio Desktop for your operating system.
4. If you cannot install RStudio, you can use Posit Cloud in a web browser or install Positron instead.

2.1.1.3 R/Positron

R and Positron are separate programs. **Install R first, then install Positron.**

1. Download and install R for Windows or macOS. Positron requires R 4.2 or later.
2. Go to the Positron download page.
3. Choose the installer for your computer. On a Snapdragon-based Windows 11 laptop, choose **Windows arm64**. If you do not have administrator access, choose the Windows **user-level** installer.
4. Install and open Positron. If prompted, select the R installation you just installed.

2.1.2 A simple modelling exercise (10 min)

What is in a question? When we see an interesting pattern and want to investigate it, we need to be able to ask questions that can be addressed with data. Let's start with a familiar example.

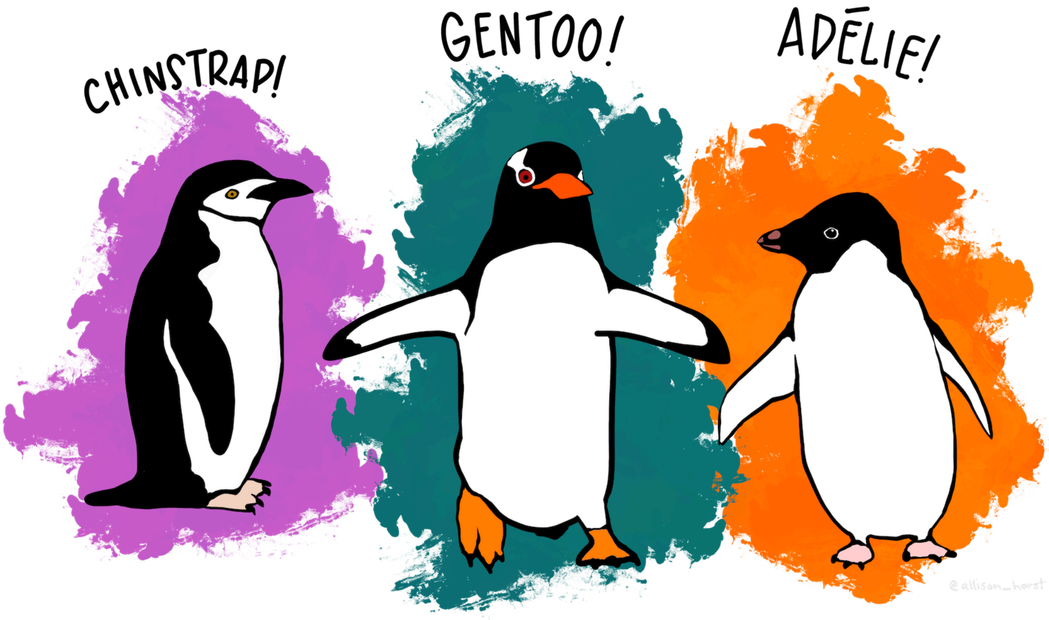


Figure 2.1: The penguin dataset. Artwork by Allison Horst.

The above are simple illustrations of the three penguin species in the `week01-penguins.csv` dataset. Suppose we were, in fact, scientists on a research expedition to the Palmer Archipelago of Antarctica, and we saw these cartoon penguins. *Imagine these are the only observations available to us: the penguins are not moving, so we can only compare their shapes and sizes, and we can see at least three species.* What interesting observations might we make about them? How could we turn those observations into a question that can be addressed with data?

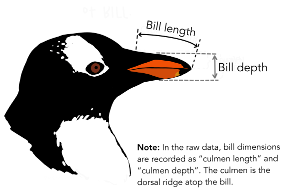


Figure 2.2: Bill dimensions in penguins. Artwork by Allison Horst.

To work with measurable data, you need to identify the **response** and **predictor** variables in a question. What is a response variable? What is a predictor variable? Write your own definitions in your lab notebook, as you will use these terms throughout the semester.

Next, we will open the `week01-penguins.csv` dataset and discuss variables, data types, relationships, and how to formulate questions. For this activity, *pretend* you are biologists who have just observed these penguins in the wild and are deciding which variables are important to measure. **We will work together to formulate a question and sketch a model of the relationship between them. No software is needed.**

2.1.3 Introduction to cheatsheets (5 min)

In the final part of the workshop, we will introduce cheatsheets, which are quick reference guides that we have developed to guide you through the software. We hope that this makes the purpose of BEDA clear: software should be the *least* of your worries.

2.2 Practical

You will now work in pairs to complete the practical exercises. Your demonstrators will roam the room and check in with you to see how you are going. If you have any questions, please ask them! **Demonstrators will check each of your exercises to ensure that you are on the right track.**

The times below are a guide. You must complete Exercise 3 in full, so ask for help early if software setup delays you.

2.2.1 Exercise: Cheatsheets (25 min)

This exercise is designed to get you familiar with the cheatsheets and how to use them. You will be asked to complete a few cheatsheets that are based on Jamovi or R. Please do not attempt other cheatsheets for now - we want to make sure you are familiar with the software today and catch technical issues early.

2.2.1.1 Task 1

1. Open Cheatsheets.
2. Choose **two cheatsheets** that you have not used before and see if you can create the outputs they describe.
3. Save those outputs. A demonstrator will check your work and give you feedback.

ⓘ **Note:** If you have never used Jamovi before, now is a good time to try it.

2.2.2 Exercise: Data types (25 min)

In this exercise you will work on the **week01-possums.xlsx** dataset. If you have not already downloaded it, do that now and open it in your chosen software.

2.2.2.1 Task 2

1. Carefully read the "Metadata" and "AllPossData" sheets in the MS Excel file. The metadata sheet describes the variables in the dataset and their units of measurement. The AllPossData sheet contains the actual data.
2. Choose six variables, including at least two continuous variables and two categorical variables. Classify each variable and record its type in your lab notebook. Some variables may have more than one reasonable classification.

3. Use those variables to identify two biologically meaningful pairs: one containing a continuous and a categorical variable, and another containing two continuous variables. You will choose one of these pairs for Task 3.

ⓘ **Note:** The same underlying trait can be recorded in different ways. For example, exact age can be treated as continuous, whereas age classes such as young, adult and old are ordinal. Binary variables are also categorical variables with exactly two levels. If you are unsure about a classification, ask a demonstrator. Data classification is a key step in the modelling process.

Data types

Data type	What it means
Continuous	A numeric measurement that can take many values
Discrete	A numeric count with whole-number values
Categorical (nominal)	A label or group with *no natural order*
Ordinal	A category with a *meaningful* order
Binary	A categorical variable with *two levels*

2.2.3 Exercise: Modelling basics (45 min)

In this exercise you will use variables from the **week01-possums.xlsx** dataset to formulate one biological question and develop one graphical model. You will choose an appropriate plot and sketch the relationship you expect to see; you do not need to analyse the data in software.

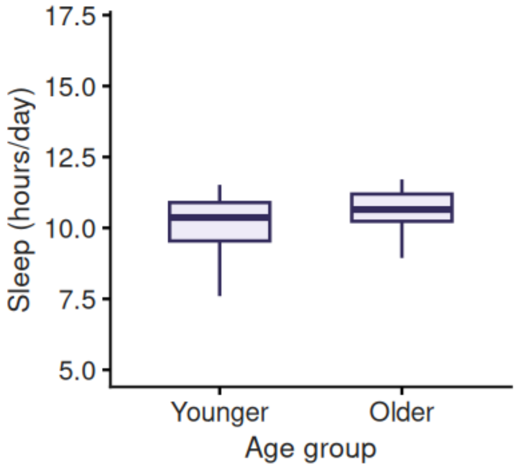
What is the point? A model is just a simple way to represent something complicated. In data analysis, a model helps us make sense of a dataset, test ideas (hypotheses), or make predictions. Models can be as simple or as fancy as you like, depending on your data and your research question.

Soon, you will use empirical models to describe relationships observed in data. We write the structure of these models using simple notation, such as $y \sim x$, where y is the response variable and x is the predictor variable. We won't fit a statistical model just yet; for now, we will use plots to represent relationships we might expect to see.

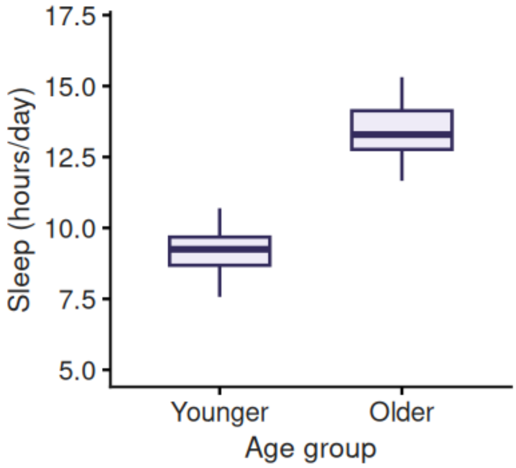
2.2.3.1 If you can plot the relationship, you can begin to model it

For now, we will use plots as simple graphical models. They are abstractions: simplified representations of relationships we expect to see between variables. These are not fitted statistical models yet; they are a way to make our expectations explicit.

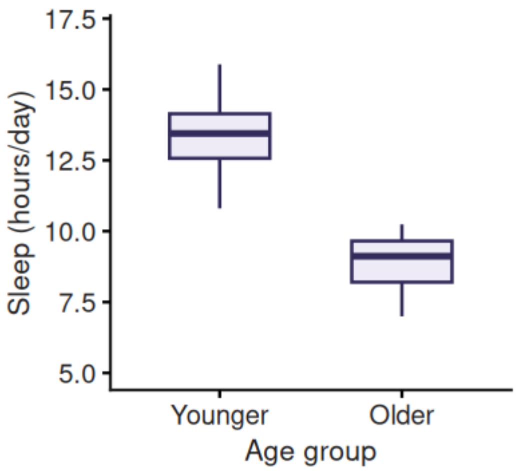
For example, you could plot the daily sleep time of dogs against their age to see if there is a pattern. **At this stage, more than one pattern may be biologically plausible. The important thing is to make your expectation explicit and be able to explain it.** Younger and older dogs might sleep for similar amounts of time, or one group might sleep longer than the other. These possibilities could look like this:



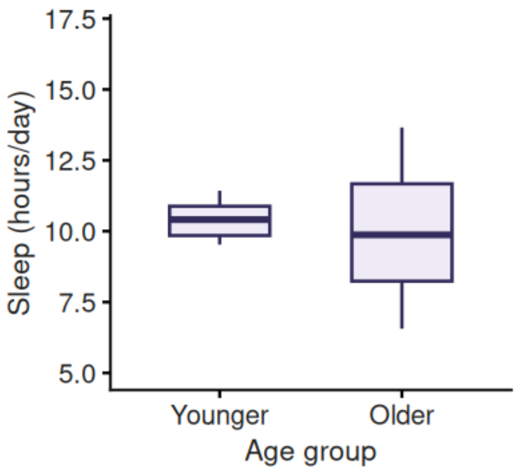
(a) Similar sleep time between age groups.



(b) Older dogs sleep longer.



(c) Younger dogs sleep longer.



(d) Greater variation in sleep time among older dogs.

Figure 2.3: Four possible patterns in daily sleep time between younger and older dogs.

Each plot represents a different possible relationship, but they all use the same response and predictor variables. The response is daily sleep time in hours, and the predictor is age group (younger or older)—a categorical variable.

Interestingly, how we consider our variables can drastically change the type of plot and model used for data analysis. For example, consider the same relationship as above, **but while planning the study, we decide to record the exact age of each dog in years**. Now, age is a continuous variable, and we can use a scatterplot to show how daily sleep time changes across the age range:

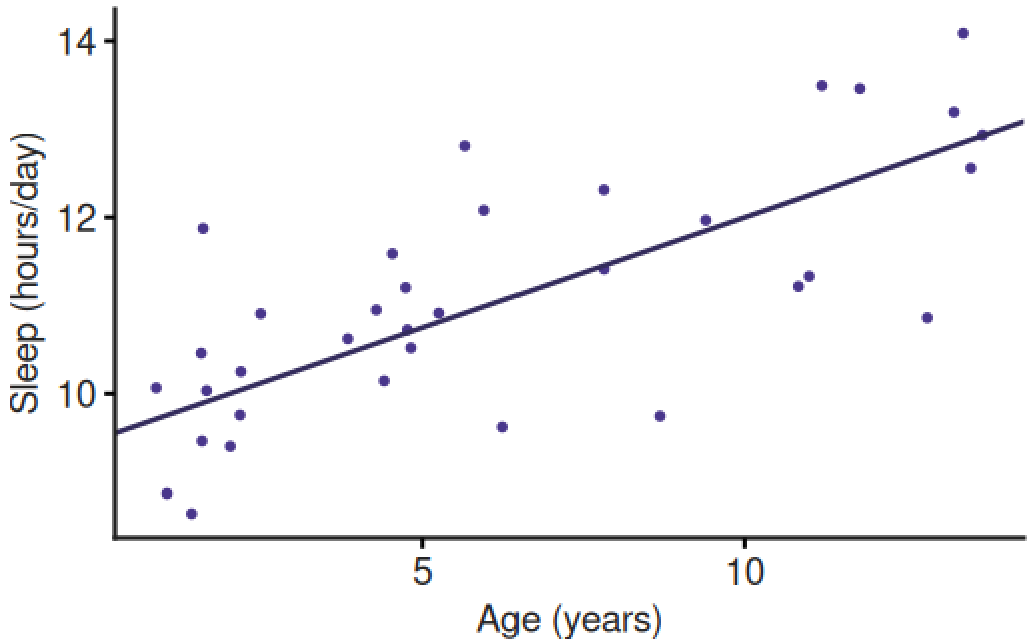


Figure 2.4: One possible relationship between daily sleep time and age in dogs.

In the scatterplot, the exact ages are retained, making it easier to see how daily sleep time changes across the age range. The boxplots and scatterplot use the same response and predictor, but treating age as categorical or continuous changes what the plot shows and how we interpret the relationship.

2.2.3.2 Task 3

Your goal is to practise thinking like a research biologist. Complete all five steps below for **one biological question and one graphical model**. You do not need to process or analyse the data. Knowing the variable types is enough to choose and sketch an appropriate plot.

1. **Choose a variable pair.** Return to the two pairs you identified in Task 2 and choose one for this task.
2. **Formulate a research question.** Choose variables that could address a simple biological question. Identify the response variable, the predictor variable and the type of each variable.
3. **Choose a plot.** Match the plot to your question and variable types:
   - **Scatterplot:** Shows the relationship between two continuous variables.
   - **Boxplot:** Compares a continuous response across the groups of a categorical predictor.
4. **Sketch your graphical model.** In your lab notebook, write your research question and draw the plot you expect to see. Label both axes and show a plausible pattern. The sketch represents your idea; it does not need to match the data.
5. **Explain your model.** State the relationship in words and explain why the plot suits your variables. Consider whether a different way of measuring or classifying one variable would change the plot you use. Discuss your reasoning with your partner or demonstrator.

Your lab notebook should contain:

- your biological question;

- the response and predictor variables, including their types;
- a labelled sketch of the plot;
- the expected relationship stated in words; and
- a brief explanation of why the plot is suitable.

You do not need to fit or test a statistical model.

2.3 End of practical notes

2.3.1 Help us collect lots of snails!

We need about 400 common garden snails for the Module 2 plant-herbivore projects! Lots of prizes in gift cards to be won. See the snail competition page for details. We have prepared **up to 17 prizes** for this competition, so check out the Canvas page and help us.

If you are interested, please collect the following from the lab before you leave:

1. A food container
2. A pair of gloves (latex or nitrile)

Taking part is optional and has no effect on your marks.

2.3.2 That is a wrap

That's it for today! If you have any questions, ask your demonstrators who are happy to help. If there is time, we will do a 10-minute Q&A session before showing the attendance QR code. Make sure to take attendance before you leave.

3

Preamble

I think one of the most exciting things about being a biologist is discovering how the world works. One skill that helps me do so, is knowing how to set up an experiment to test my ideas about the world. You may need to overcome that initial learning hump, but if you persist, you will have a skill at your fingertips that will stand you in really good stead for the future. It doesn't matter whether you intend to go on in research or not - knowing what makes good research and good evidence is important in so many walks of life.

My message is: whether you are doing this unit just because you have to, or because you are interested – take heart! I never did anything much more complicated than a t-test even during my PhD. But now I LOVE designing experiments and grappling with data. It can be frustrating at first, especially if you are time-poor and just want to get the job done. It can be infuriating. It can also be fun (yes – fun!). Most of all it's incredibly satisfying and extraordinarily useful. So, persist, try your best, and ask for help if you need it.

Please remember we want you to learn this stuff.

No question you ask is EVER silly! So ask for help any time. We *do* expect you to have a go at telling us what you *think* the answer is. This helps us understand what you get and what you don't get, and so helps us help you.

We don't judge you by the questions you ask, but by what you end up learning.

4

Module 2 timeline

Equipment and technical help: Please refer to the BIOL2022 Canvas site for the current technical officer's contact details and arrangements. When requesting help or equipment, include your project name, group name and practical time. Clare, technical staff, and demonstrators will be available during practical sessions to discuss expectations and project plans. Your group should aim to be self-sufficient outside practical times. For some projects, you will be able to take gear home straight away to run a pilot study; for others, plan now and pick up gear in consultation with technical staff.

Submission deadline **Friday 25 September 2026 at 23:59**

Week	In your timetabled session	Between practical sessions and in your own time
Weeks 2–3 10–23 August	**Sign up to a project group for Module 2 in your Prac Session.** Everyone in your group must attend the same prac session.	**Choose a direction:** Review the project information. Make sure all group members are in the correct Canvas group.
Week 4 24–30 August	**Plan and pilot:** Discuss and plan the project, then form a hypothesis and predictions. Design a simple main experiment you can analyse with familiar statistics. Sketch the expected graph and decide which statistics to use, and why. Draft data sheets and work out the equipment you need. Pick up gear for the pilot if needed. For some projects, you will be able to take gear home straight away to run a pilot study. For others, plan now and pick up gear in consultation with technical staff. Everyone contributes to the planning.	**Run the pilot:** Plan and run it before Week 5. In your own time, continue pilot work, read the literature, and begin report preparation.
Week 5 31 August–6 September	**Refine and run:** Work with your group; finalise the main design in light of the pilot and literature. Clarify the response and explanatory variables and the analysis. Pick up any remaining gear. Run the main experiment and collect and collate the data as a group.	**Collate the data:** Keep one shared Excel/Google Sheets workbook with metadata and a clean results array. Finish data collation before Week 7 if possible.
Week 6 7–13 September	**Use the period for project work.** There is no structured practical this week. Use the time to finish data collection, troubleshoot, and organise the group dataset.	Make sure everyone contributes to collecting and organising the data.
Week 7 14–20 September	**Analyse and interpret:** Work with your group; finalise data collation and metadata (no more than 30 minutes). Metadata should cover group members, dates, sites, response variables and units, explanatory variables (treatments / levels / continuous variables), and column explanations. Include anything that would help you understand your project and data	**Begin the report:** Finish your analyses and graphs. Draft the Introduction, Methods, and Results using the report instructions and rubric. Continue your data analysis if you have yet to complete it. Start drafting your report: begin with a framework for the Introduction,

Week	In your timetabled session	Between practical sessions and in your own time
	if you were to publish it for the world to see and understand, such as what the columns in your data sheet mean. Spend the remaining time (at least an hour) learning to plot, analyse, and interpret in Excel and R or SPSS. Collaboration is fine, but each student analyses independently and decides on plots and statistical output.	Materials and Methods, and Results. You could leave the Discussion until you have finalised your Results.
Deadline	**Finalise submission:** Use the report-writing guidance; finalise analyses, statistical output, and figures. Prepare the group Excel file with metadata, a clean data array ready to import into a statistics package, and supporting worksheets. Name the file Day-Time-Project (e.g. WED2-4pmProj4A). Each student works on their own report, results, graphs, and presentation. Each student should complete their own Results section and graphs, decide what to include in their individual report, and decide how to present the data and associated statistics.	**Submit both components.** One group member submits Report 1: Group data submission (the group data file). Every student submits Report 1 (individual report). Both are due Friday 25 September 2026 at 23:59.

5

Report 1 – Projects

FOR ALL PROJECTS – keep your experiment simple. You should be able to design an experiment that could be analysed using one of the following:

- a paired t-test (Week 3),
- an independent t-test (Week 3),
- 1-way ANOVA, 2-way ANOVA (Week 4),
- correlation or linear regression (Week 4),
- Chi-Square analysis (if necessary) or
- logistic regression (if necessary, Weeks 6-7)
- GLM (which covers a lot of the analyses above) and GLMM are fine **if applied correctly**.
- You will be rewarded for doing a simple appropriate analysis correctly.
- **You will be penalised for doing a complicated analysis wrong.**

5.1 Summary

ID	Project and format	Things to consider
P01	Tomato herbivory after leaf damagePlant–snail experiment	Hands-on, controlled experiment using plants and snails; includes animal handling.
P02	Vegetable herbivory after leaf damagePlant–snail experiment	Hands-on, controlled experiment using plants and snails; includes animal handling and comparison between plant species.
P03	Ant food preferencesAnt-foraging experiment	Field-based, likely in backyards or another controlled outdoor setting. Your group will set up food treatments and observe ant recruitment.
P04	Ant foraging costs and benefitsAnt-foraging experiment	Field-based, likely in backyards or another controlled outdoor setting. Your group will manipulate access to food and observe ant recruitment.
P05	Wheat seed germination and growthGermination experiment	Requires carefully controlled conditions and regular care of plants over time.

ID	Project and format	Things to consider
P06	Native grass, wheat or horticultural seed germinationGermination experiment	Requires carefully controlled conditions and regular care of plants over time. You can switch to P05 if germination fails.
P07	Bird activity across time of dayBird survey	Outdoor activity across a broader geographic area. Animal ethics conditions apply: observe birds only; do not handle or disturb them.
P08	Bird use of urban water bodiesBird survey	Outdoor activity across multiple water bodies or sites. Animal ethics conditions apply: observe birds only; do not handle or disturb them.

5.2 Projects

5.2.0.1 Project P01 – Plant-herbivore interactions: Does previous damage to tomato plants affect their vulnerability to subsequent herbivory? Does the amount of previous damage matter?

Background: Many plants have plant secondary metabolites (PSMs), which often play a role in defence against herbivores. In some plant species, PSMs can be induced (levels increase) after damage from herbivores chewing on leaves. This change may take place within a day (but check previous studies), affecting subsequent leaf damage by herbivores, and result in less subsequent damage. Is the evidence consistent with this idea?

Plants and herbivores: this project will likely use tomato seedlings and common garden snails. Your group may need to collect at least some snails yourself.

Variation: If you want to answer a more complicated question, you could ask whether the amount of initial damage (a little versus lots?) affects the outcome.

NOTE: You will not measure PSMs. Rather, they provide a potential mechanism to explain any difference in subsequent herbivory.

5.2.0.2 Project P02 – Plant-herbivore interactions: Does previous damage affect vulnerability of vegetables to subsequent herbivory? Does it depend on plant species?

Background: Many plants have plant secondary metabolites (PSMs), which often play a role in defence against herbivores. In some plant species, PSMs can be induced (levels increase) after damage from herbivores chewing on leaves. This change may take place within a day (but check previous studies), affecting subsequent leaf damage by herbivores, and result in less subsequent damage. Is the evidence consistent with this idea? Does it depend on plant species?

Plants and herbivores: this project will likely use tomato and cucumber seedlings and common garden snails. Your group may need to collect at least some snails yourself.

NOTE: You will not measure PSMs. Rather, they provide a potential mechanism to explain any difference in subsequent herbivory.

5.2.0.3 Project P03 – Foraging strategies in relation to food type: Do ants prefer some food over others?

Background: Ants spend a lot of their time searching for food, but the food they find varies in its nutritional and energetic value. Do they just forage for any foods or do they show preferences? Honey and tuna, for example, are very different foods. Do ants care? Honey is a high-carbohydrate low-protein food, while tuna is high protein, low carbs; although there are also lots of other differences between these foods.

Scout ants go out and look for food, then return to the nest and recruit others to help them. We can understand something about the requirements ants have for different foods by looking at how quickly and/or how many ants are recruited to a food source once it has been located by the scout ant.

5.2.0.4 Project P04 – Foraging strategies in relation to costs and benefits: Do ants seek food that is easier to access?

Background: Ants spend a lot of time searching for food, but the effort needed to harvest a given food could affect how they value it. For example, food that is difficult or slow to access (e.g. surrounded by complicated 3-D local habitat or with narrow access points slowing ant traffic) may be less preferred than the same food that is easy to access.

Variation: IF you want to answer a more complicated question, you could ask whether ants balance the effort of getting food against the value of that food. Does food that is hard to access need to be higher in value (e.g. more concentrated) than food that is easy and less energetically costly to access?

Scout ants go out and look for food, then return to the nest and recruit others to help them. We can understand something about the requirements ants have for different foods by looking at how quickly and/or how many ants are recruited to a food source once it has been located by the scout ant.

5.2.0.5 Project P05 – Germination and growth of wheat in relation to water availability

Background: Growing crops such as wheat is crucial for food security. In Australia, our rainfall is highly variable, and this can be daunting for farmers wondering whether to sow seed and whether it will germinate if they do. White wheat varieties in Australia can germinate under a range of temperatures (12 to 25oC), but whether the initial rainfall is light (e.g. shower) versus heavy (e.g. soaking) may affect whether seeds germinate and/or seedlings grow and/or survive.

Understanding these relationships is one step in understanding the conditions required to help ensure our food security.

5.2.0.6 Project P06 – Germination of seeds of native grass OR wheat OR a horticultural plant (flower or veggie) in relation to water availability

Background: Germination of **native plants** is triggered by a range of factors in Australia. Some plants need heat initially, others simply need water. If we can, we will provide you with seed from a native plant that germinates quickly under the right conditions. How often it rains may affect whether seeds germinate, in turn affecting recruitment and regeneration. Understanding the relationship between rain (watering) frequency and probability of seeds germinating helps us understand how climate influences the dynamics of our grasslands and use of native plants in agriculture.

NOTE: In your pilot study or main trial, if you get no germination after 7 days, your group can choose to switch to wheat (Project P05) or seeds of a horticultural plant to ask the same question, because these germinate quickly.

Germination of wheat is crucial for food security (see Project P05).

Germination of horticultural plants is crucial for the cut flower or veggie industries; no germination, no plants, no business! How often it rains or how often seeds are watered may affect whether or how quickly and/or how many seeds germinate – but you need to find this out.

5.2.0.7 Project P07 – Are birds active at different times of day?

Background: Birds may be active at different times of day for a variety of reasons. As a first step in understanding bird abundance and diversity we need to be able to measure it and know whether when we measure it makes a difference. If you stand in the same spot and count birds, does the outcome depend on when you did it? Are there other factors you need to control or at least take into account?

NOTE: Animal ethics and bird safety conditions apply (see below).

5.2.0.8 Project P08 – Use of urban water bodies by birds: Do characteristics of urban water bodies influence which birds use them?

Background: Bird diversity within cities is affected by many factors, including availability of food, sites for nesting and shelter from predators. Water bodies, such as ponds and lakes, in parks and gardens can attract native (and introduced) birds. The Sydney Council, for example, has upgraded and revegetated the ponds in Sydney Park, just south of Sydney Uni, to encourage water birds. To maximise bird diversity, as well as simply maximise bird numbers, it is important to understand the relationship between characteristics of different water bodies and the diversity of birds they support.

NOTE: Animal ethics and bird safety conditions apply (see below).

5.3 Animal Ethics requirements for Projects on birds

If you sign up to these projects, you must follow the animal ethics requirements as approved by the University of Sydney Animal Ethics Committee:

5.3.1 Bird safety and biosecurity

Do not touch, handle, move, clean up, or transport birds or other wildlife. In the very unlikely event that you find injured, sick or dead birds while undertaking this project, keep your distance, record what you observe, and follow the guidance below.

- **Sick or dead birds:** at the time of writing, H5 bird flu has been detected in a low but increasing number of wild birds along the coast of Australia. Even though this project does NOT involve any handling of birds, you must avoid touching any sick or dead birds, and any sighted sick or dead birds need to be reported to the Emergency Animal Disease Hotline on **1800 675 888** (as outlined in the NSW Department of Primary Industries and Regional Development guidance).
- **Injured birds:** do not handle, move or transport the bird yourself. Seek advice from Clare McArthur (clare.mcarthur@sydney.edu.au, subject heading: "BIOL2022 URGENT-bird project")

or another academic in the unit of study, who will advise you on contacting a wildlife carer or wildlife rescue organisation.

- In every case, notify Clare McArthur by email within 24 hours, summarising what has occurred (this will be reported as an adverse event to the Animal Ethics Committee even if you played no part in the injury).

Project "BIOL2022 Ecology teaching-bird surveys". Project Number: 2026/AE000123.

You may collect data for up to 10 visits per site over a 2 to 4 week period. For the bird survey, you may observe birds, count them and identify each to species if you can, using binoculars where needed. You may observe birds at a site for up to ~ one hour per visit, but must not interfere with nor handle the birds.

At a given site, the surveyor (1 to 5 students per site) will find a suitable location to observe birds. They will stand or sit still for 15 to 30 minutes per location and count the birds they can see in their site. They may need to move to up to five other locations per site if they have difficulty identifying the birds from afar. The number of each species will be recorded with reference to bird guides, and a total of about one hour will be spent at each site per observation period.

Depending on your project, you may quantify other data, such as vegetation assessments and size of any habitat types including water bodies, at each site, enabling a test of the factors affecting bird abundance and species diversity.

Observers will remain far enough from birds that the birds are not visibly affected by them (i.e. do not alter their behaviour, either by moving away or by altering what they are doing). This avoids any potential impact of the observer on time or energy budgets of the birds (as described in Bateman et al. 2011).

Reference: Bateman, P. W., and P. A. Fleming. 2011. Who are you looking at? Hadeda ibises use direction of gaze, head orientation and approach speed in their risk assessment of a potential predator. Journal of Zoology 285:316-323.

5.3.2 Conditions of bird project approval:

1. You must use the approved monitoring sheet "2026 BIOL2022 Bird survey monitoring sheet" (pdf or xlsx), which will be provided to you. You may add additional columns or have additional sheets, as you decide, appropriate to your study.
2. You must take photos of the fieldwork sites and the observers watching the birds (i.e. to see the relative locations of the observers to the birds).
3. By the end of Week 9, you must post on Canvas:

    a. A complete, legible copy of the final monitoring sheets from (1)
    b. A summary of the total number of (i) native birds, (ii) invasive birds hence (iii) all birds you counted at each site and across all sites over the course of your project, and a list of the sites you monitored, with GPS coordinates.
    c. Clearly and informatively labelled photos from (2).

6

Report 1 – Instructions

6.1 About

- The project written report is the individual component of this assessment It is submitted via the Canvas assignment called **Report 1**.
- The other component of this assignment is the data file, which should be submitted by one group member via the Canvas assignment called **Report 1: Group data submission**.
- **Date due**: Friday 25 September 2026 at 23:59
- **Late penalty**: you will incur a penalty of 5% (5 points out of 100) for each 24 hours or part thereof after the deadline (weekends and public holidays included), unless you have Special Consideration.
- **Length penalty**: you will incur a penalty of 5% (5 points out of 100; up to 30 points) for each page or part thereof over the page limit.

6.2 A note about plagiarism

Do not plagiarise or you will face penalties. Check the academic integrity website for access to:

- the policies and codes covering academic honesty and conduct at the University.
- a link to the AHEM (Academic Honesty Education Module); complete the module if you haven't already done so. You cannot submit your report if you have not completed this module.
- If you are still unsure about how to avoid plagiarism, check with Clare McArthur well before submitting your report.
- Your report will be checked for plagiarism using software.

6.3 Submitting your individual report

- **Submitting your anonymous report for marking**:
  - ONE electronic document through Canvas. As this needs to be anonymous, do not include your name anywhere in the report.
  - Use a filename of the form "Day-Time-Project-Your SID", e.g. "WED2-4pmProj4A-Your SID".
- **Using your report to help other students**: so we can provide useful feedback using real examples, your report or parts of it may be used as (anonymous) examples for current and future students. If you wish to opt out of this, please make a statement to this effect on your report.
- **Page limit**: maximum 10 pages double-spaced, including everything (i.e. abstract, references, your statement about use of AI, and everything else).
- **References**: minimum of 10 key references from the primary scientific literature, probably a mix of key recent and older papers. They must be relevant, demonstrating that you have read about the topic comprehensively. You will use them to show how you understand what people

have already done and how your work fits into and/or extends what is already known, and that you have used the references to help understand what you have found.

6.4 Requirements for your report

- You will be marked according to the criteria in the marking rubric for this module. Read and USE this rubric as a check-list before, during and after writing your report and before you submit it.
- Use minimum of 12 pt font and no less than 2 cm margins all around.
- You must write all your report yourself, in your own words, on the experimental research project you have done. **If you use AI, you must follow the university guidelines and you must provide a statement on its use (or if not, state that instead).**
- Your report should follow the format and general content of a real scientific article. To get a feel for what we expect and how to present each part of your report, read the published literature in your subject area.
- Choose a relevant journal style and follow it (but don't use "newsprint" columns); e.g. note that figure headings are below and table headings are above the figures/tables.
- We expect information from the primary literature; i.e. academic peer-reviewed sources, particularly scientific journals. Avoid websites: they usually lack quality control, and they come and go. I recommend that you use:
  - Web of Science (and Google Scholar) as your main portal for accessing journal articles, searched using key words and
  - EndNote or Mendeley for organising your references (but it is not essential). Consult the Library for short courses on how to use these if you don't know already.
- **If you are uncertain about anything, seek help early from the teaching staff in class.**

6.5 Clare's tips on writing your report

A great reference on how to write well:

Mensh B, Kording K (2017) Ten simple rules for structuring papers. PLoS Comp. Biol. 13. doi: 10.1371/journal.pcbi.1005619

START EARLY, START EARLY, START EARLY: you can start to draft your introduction as early as Week 5, once you have read the literature on the topic and have a feel for what your experiment will cover. Don't try to make it perfect – just get some words down. You can also start to draft your Materials and Methods section once you have started your experiment and before Week 7. You should be able to draft your result section after Week 7 and then finalise your whole report, including your discussion, in Week 8 and before submitting at the end of Week 8.

Title: keep it brief, **relevant**, informative and interesting.

6.5.1 Abstract

- Brief abstract (max. 120 words), including (1) background. (providing the context for your problem/project), (2) hypotheses or aims, (3) what you did, (4) what you found, (5) what it means, hence could be five sentences.
- Do NOT include references or details of statistical results (e.g. no F or P values).

6.5.2 Introduction

- What do we know already? Make sure it leads logically to your aim(s). Provide the context - previous studies on the subject, using published (mainly) primary scientific literature, NOT websites. Integrate information clearly and concisely without loss of original meaning, to provide a logical, well-interpreted review of existing research and other information.

- Aim of the experiment and biological hypotheses (IF…) and predictions (THEN…).
- Why is the study important and/or interesting.
- Study system – species, location of experiment (brief).
- Definitions of scientific terms (if needed).

6.5.3 Materials and Methods

- Pilot study (if needed, keep it brief).
- Set-up of experiment.
- Sampling design.
- Data / variables recorded and how.
- Statistical analyses used, including which are dependent/independent variable(s) (if applicable), unit of replication, actual test(s) & why they are appropriate, statistical package(s).

6.5.4 Results

- Written statements describing results (data and stats). Refer to tables and figures as needed. **Do NOT simply provide the tables and figures without supporting text.**
- Summary presentation of data (no raw data) in text & as figures or tables as appropriate. Use figures to illustrate patterns, use tables for large amounts of data where detail is important. Indicate sample size and errors, and note whether the latter are standard errors or standard deviations… Format consistently. Make sure decimal places reflect your real capability to measure (e.g. can you really measure a 1 kg plant to 6 decimal places?)
- Results of statistical analyses presented either in text, on figures, or in tables.

Important tips:

1. **Stats and AI:** You are NOT allowed to upload your data set to any AI platform. If you need help to work out what stats is appropriate and WHY, you can use AI (as a complement to the lectures). BUT you MUST analyse your data yourself. Your tutors and demonstrators will ask each of you to show them your statistical analysis in the prac class. If you cannot do this, you will need to work with their guidance until you are able to do it.
2. Know what stats matter from the output – there are lots of them!
   - Only **some/key stats** need to be included in your report
   - **Do not copy and paste stats output from your stats package**
   - DO type the key stats results up yourself and only include what matters – that way we know you know what matters
3. Check assumptions, if any are associated with your stats analysis:
   - In your Methods section of your report: say what/whether you have checked, whether they were met, if not, what did you do?
   - in your report DO NOT present normality plots or plots of residuals
4. Prepare your formal graph(s) for your report
   - You may present the graph produced by your stats package

6.5.5 Discussion

- Main conclusions or findings
- Interpret results – what do they mean? How do they relate to original aims?
- Compare with and discuss in context of previously published work.
- "Limitations": every single study has some context and defined approach or methods. This does NOT mean it is necessarily flawed. Take a positive approach to your research and how you word what you have done. If you do want to mention/discuss these constraints, do not reduce your

discussion to a lengthy apology. You do not need to clutch at straws trying to find something wrong with your project so you can write about "limitations" section.

- **What to do if your group made a genuine error?** Even though I would hope your group had clearly discussed your experimental design & analysis with me/tutor/demonstrator in Weeks 4 and 5 to ensure your design was robust, if your group did make a genuine error by mistake/lack of understanding, then yes – say *what* was wrong. But I want you to learn from mistakes, so you will not lose marks provided you say *why* it was wrong and *how* you would rectify it if you had the chance to do the study again (you won't earn as many marks as if you had designed things well in the first place).
- Future studies – what would you do next to extend our understanding of the subject and/or improve what you did? Keep this brief.

6.5.6 General

- Write in PLAIN English:
  - Be succinct and avoid trying to sound "scientific".
  - read it out aloud, if it sounds pompous it is, and it should NOT be.
- Correct presentation (format/length).
- References used and cited correctly. You are expected to use at least 10 relevant references from the primary literature.

6.6 More tips on writing your report

The following is a summary written by Clare's Lab Group of PhD and Honours students after critically discussing two papers — not for content — but for how they were written.

6.6.1 Structuring Research Papers: by Clare's Lab Group

"Rules" based on Mensh B, Kording K (2017) Ten simple rules for structuring papers. PLoS Comput Biol 13(9): e1005619. https://doi.org/10.1371/journal.pcbi.1005619

6.6.1.1 Papers discussed:

Cline, B. B. and Hunter, M. L. (2014), Different open-canopy vegetation types affect matrix permeability for a dispersing forest amphibian. J Appl Ecol, 51: 319-329. doi:10.1111/1365-2664.12197

Palmer, M. S., Fieberg, J. , Swanson, A. , Kosmala, M. and Packer, C. (2017), A 'dynamic' landscape of fear: prey responses to spatiotemporal variations in predation risk across the lunar cycle. Ecol Lett, 20: 1364-1373. doi:10.1111/ele.12832

6.6.2 General notes

- Do write for your reader, do not write for yourself.
- If you read the first sentence of every paragraph, it should tell a story.
- The order of your ideas in each section should be the same; i.e. consistent order in intro, MM, R and probably D of your aims, methods for each aim (provided that works), results and discussion of your results associated with each aim.
- Subheadings can make sections easier to read and understand the flow of ideas.
- Avoid zigzagging ideas, i.e. you should never need to say "as mentioned above".
- Use consistent wording – DO NOT use a variety of words to mean the same thing because it is confusing for readers. They will not know if you are referring to one thing or several. DO: pick one name for a variable and stick to it.

- Take the reader by the hand. DO NOT assume they know what you know. DO lead them through your work and explain things as you go along. Justify your decisions. Think like a reviewer: what will they want to know? Pre-empt it by being transparent, explaining what you decided to do and why.

6.6.2.1 Introduction

- There will generally be more references in the intro than the discussion. The focus of the intro is on: what do we know already?
- References are used to provide background of the research area – do not just list references on a topic. Instead, use them to summarise what we know and understand as a result of these studies. Make it clear what we know from (a) empirical evidence (with examples, i.e. do we know it for lots of individuals/species/contexts/ecosystems or whatever? which? Just one? which?) OR (b) what we think we know (conceptual hypotheses).

6.6.2.2 Suggested Paragraph 1:

- Introduce the broad topic.
- Describe what research has been done on this topic.

6.6.2.3 Suggested Paragraph 2:

- Start to narrow down what area in this topic your paper will be covering
  - Introduce gaps in knowledge.
  - Possibly describe issues with previous studies (that your paper will address).
- This information can show the importance of your paper.

6.6.2.4 Suggested Paragraph 3 to 5:

- Specifically describe what your paper will investigate.
- State your aims and hypotheses/predictions. These should follow and funnel down from your background. If you put your hand over your aims, you should be able to guess them from the background you provide above.
- Briefly describe how you investigated the topic.
- State any predictions with information to backup these predictions.

6.6.2.5 Discussion

- The focus is now on what you found, what it means and so how you have advanced our state of knowledge & understanding as a result of what you have done.
- Therefore: generally fewer references in the discussion than the introduction.
- References should be used to support your idea or relate your conclusions/results to previous studies.
- The following paragraph order makes sense but you can obviously increase or decrease the number of paragraphs depending on the context and size of your study.

6.6.2.6 Suggested Paragraph 1:

- State any broad results that you found.
- Provide evidence from your results that enabled you to come to that conclusion.
- Compare the result to any predictions you made or previous studies.

6.6.2.7 Suggested Paragraph 2 to 4:

- State more detailed results (e.g. species specific results if dealing with multiple species).
- Provide evidence from your results that enabled you to come to that conclusion.
- Compare the result to any predictions made or previous studies.

6.6.2.8 Suggested Paragraph 5:

- Limitations of your study
  - No study can answer everything and every study is limited in some way in what is achieves! Don't worry! That's life!
  - Don't dwell on this or go too much in depth – especially do not be apologetic.
  - Be positive: just outline the boundaries of the conclusions that can be made with your results.
- Suggested direction of future studies i.e. what could be done next to move forward in our understanding and/or resolving any unanswered questions given what you have now discovered – again keep this brief.

6.6.2.9 Suggested Paragraph 6:

- What are the implications of your study for your field of research
- Describe what this information can affect or be used for (e.g. management decisions).

6.6.2.10 Suggested Paragraph 7:

- You do not need a concluding paragraph – but if you do, it is not just a summary.
- It should be more big picture stuff but take care: it should not state something that could have been said whether or not you had done your study. It must rest on your study.

7

Weeks 4–8 practicals: resources

7.1 Useful documents for Report 1

- Here are two examples: template1 and template2 for Excel group data collection. Generate and complete your group's own Excel file (could be from a Google Sheets file) **before** Week 8 prac.
- Here is a guide to choosing appropriate stats and a guide to using the stats program SPSS.
- Practice Excel file for ANOVA or Randomised Block design/GLM and associated paper.
- Practice Excel file for linear regression (or correlation) and associated paper.
- Practice Excel file for logistic regression and associated paper.
- And an interesting article called "The importance of stupidity in scientific research": directed at PhD students but also, I think, relevant to our Module projects where you take a step into the unknown. Hope you enjoy it.

7.2 BIRD GROUPS ONLY

! Important Animal Ethics information

Please read the 2026 Animal Research Authority (ARA) and use the 2026 Bird survey monitoring sheet (PDF) for your data collection. For spreadsheet editing, use the editable monitoring-sheet template (XLSX).

As part of the Animal Ethics requirements each bird group must submit:

1. at least one example of this completed monitoring sheet, along with
2. a pdf document summarising (a) the sites visited (names with suburb, and if possible GPS coords and site photos), and (b) the total number of (i) native birds and (ii) invasive birds you estimate you have counted across all your sites.

8

Report 1 – Rubric

The Report 1 marking rubric will be released soon on Canvas. Please check Canvas for the current rubric before you begin preparing your report.